Dr. Ponnappa Kechanda Prasanna

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DOB: 18 November 1995
Kodava Samaja Road, Ammathi Village and Post,
Virajpet, Kodagu, Karnataka, India - 571211

Education

June 21 - Jan 26
Ph. D. - Physics
MATES LAB (Supervisor : Dr. Sudip Chakraborty)
Harish-Chandra Research Institute (HRI) Prayajgraj, Uttar Pradesh (India) - 211019
(An aided instute of the DAE, A C.I. of Homi Bhabha National Institute)
Thesis title: *Theoretical Study of Lower Dimensional Materials for Energy Scavenging.*

Oct 19 – May 21
Ph. D. - Physics (Group moved to HRI)
MATES LAB (Supervisor : Dr. Sudip Chakraborty)
Indian Institute of Technology, Indore (IITI), Madhya Pradesh (India)-452020
Thesis title: *Two Dimensional Materials for energy harvesting.*

Aug 16 - May 18
M.Sc. - Physics (7.526 CGPA)
Specialization: *Condensed Matter Physics.*
Manasagangotri - Mysore, Karnataka (India)-570006
University of Mysore
Thesis title: *Simulation of Diffraction from Rectangular Apertures*
(Supervisor: M. A. Shridhar)

Aug 13 - May 16
B.Sc. (79.58%)
Specialization: *Physics, Mathematics, Computer Science.*
SBRR Mahajana FGC, Mysore, Karnataka (India) - 570012
University of Mysore

Competative Exam / Fellowship

Infosys Fellowship — Aug 2024 - Aug 2025

JOINT CSIR - UGC NET EXAMINATION — **AIR 94** (June 2019)
Physics

GATE — **AIR 382, SCORE 559** (March 2019)
Physics

JEST — **AIR 550, 92.46 percentile** (March 2019)
Physics

KSET — March 2019
Physics

Teaching Experience

Jan 24 - May 24
Teaching Assistant HRI
Tutor for Advanced Numerical Analysis Course for First year M.Sc and Ph.D

Nov 19
Teaching Assistant IITI
Tutor for Advanced Numerical Analysis Course for Second year M.Sc

Aug 19 - Sep 19
Guest Lecturer, GFGC Kuvempunagar, Mysore, Karnataka (India) - 570023
Handled first- and second-year lab experiments, teaching Modern Physics to first-year B.Sc. students and Electronics to second-year B.Sc. students.

Research Interests

Density Functional Theory(DFT), Two Dimensional Materials, Photovoltaics, Photoelectrocatalysis, Excitons

Skills

Languages	Reading, writing and speaking ability for English, Kannada, Hindi & Kodava Thakk.
Coding	C, C^{++}, Fortran, Python, Perl, GNU Octave,
First-principle calculations	Vienna Ab initio Simulation Package(VASP) and Quantum Espresso (QE) and Yambo

Research Publications

Journal Articles

1. **P. K. Prasanna**, S. Bhattacharya, and S. Chakraborty, "Interfacial configuration impact on rashba spin splitting, excitonic radiative lifetime and band edge alignment in janus monolayers based heterostructures," *Journal of Materials Chemistry A*, vol. 13, pp. 32 232–32 245, 2025. DOI: https:/doi.org/10.1039/D5TA04076K.
2. **P. K. Prasanna** and S. Chakraborty, "Emerging chalcogenide based van der waals heterostructures for ultrathin excitonic solar cells with enhanced photo-conversion efficiency," *Journal of Materials Chemistry A*, vol. 13, pp. 7949–7961, 2025. DOI: https:/doi.org/10.1039/D4TA08172B.
3. **P. K. Prasanna** and S. Chakraborty, "Threefold impact on reaction coordinate mapping of zirconium based tri-chalcogenide pseudo-monolayers as emerging ultrathin catalysts for water splitting," *Advanced Theory and Simulations*, vol. 8, p. 2 400 816, 2024. DOI: https:/doi.org/10.1002/adts.202400816.
4. R. Sadhukhan, A. Kumar, **P. K. Prasanna**, A. Guha, R. Arenal, S. Chakraborty, and T. N. Narayanan, "Ultra-low-loaded platinum bonded hexagonal boron nitride as stable electrocatalyst for hydrogen generation," *ACS Applied Materials & Interfaces*, vol. 16, no. 13, pp. 17 063–17 064, 2024. DOI: https:/doi.org/10.1021/acsami.4c04036.
5. D. Singh, S. Singh, **P. K. Prasanna**, R. K. Rai, P. Chirawatkul, S. Chakraborty, M. Fichtner, and P. Barpanda, "Zinc-substituted cobalt phosphate ZnCo2(PO4)2 as a bifunctional electrocatalyst," *Inorganic Chemistry*, vol. 62, no. 31, pp. 12 345–12 355, 2023. DOI: https:/10.1021/acs.inorgchem.3c01367.
6. P. F. M. Elango, S. Kabir, **P. K. Prasanna**, M. X. Low, M. Yang, S. Chakraborty, S. Walia, S. Sriram, and M. Bhaskaran, "Vanadium dioxide-based miniaturized thermal sensors: Humidity effects on phase change and sensitivity," *ACS Applied Electronic Materials*, vol. 4, no. 11, pp. 5456–5467, 2022. DOI: https:/doi.org/10.1021/acsaelm.2c01110.
7. S. Mondal, S. Sarkar, D. Bagchi, T. Das, R. Das, A. K. Singh, **P. K. Prasanna**, C. Vinod, S. Chakraborty, and S. C. Peter, "Morphology-tuned Pt3Ge accelerates water dissociation to industrial-standard hydrogen production over a wide ph range," *Advanced Materials*, vol. 34, no. 30, p. 2 202 294, 2022. DOI: https:/doi.org/10.1002/adma.202202294.
8. A. G. Saraswathy Vilasam, **P. K. Prasanna**, X. Yuan, Z. Azimi, F. Kremer, C. Jagadish, S. Chakraborty, and H. H. Tan, "Epitaxial growth of gaas nanowires on synthetic mica by metal–organic chemical vapor deposition," *ACS Applied Materials & Interfaces*, vol. 14, no. 2, pp. 3395–3403, 2022. DOI: https:/doi.org/10.1021/acsami.1c19236.

9. N. Sethulakshmi, S. Nellaiappan, **P. K. Prasanna**, T. Das, S. Irusta, S. Chakraborty, and S. Sharma, "Nanocoral architecture for enhanced hydrazine assisted water oxidation: Insight from experiment and theory," *Journal of Electroanalytical Chemistry*, vol. 922, p. 116 776, 2022. DOI: https:/doi.org/10.1016/j.jelechem.2022.116776.

10. S. Parveen, **P. K. Prasanna**, S. Chakraborty, and P. Giri, "Stable deep blue emission with unity quantum yield in organic–inorganic halide perovskite 2d nanosheets doped with cerium and terbium at high concentrations," *Journal of Materials Chemistry C*, vol. 9, no. 7, pp. 2437–2454, 2021. DOI: https:/doi.org/10.1039/D0TC04937A.

Under Revision/To be Submitted

1. **P. K. Prasanna** and S. Chakraborty, "Many-body pertubration theory based probing of inter and intra-layer exciton delocalization in indium-based chalcogenide heterostructures,"

Workshops & Conferences

15-26 Jul 2024 — **Poster Presentation** "Emerging Van der Waals Heterostructures for Ultrathin Solar Cells: Implication of Type-II Band Edge Alignment on Photo-Conversion Efficiency"
Engineered 2D Quantum Materials
ICTS Banglore, India

22-26 Apr 2024 — **Oral presentation** "Emerging Van der Waals Heterostructures for Ultrathin Solar Cells: Implication of Type-II Band Edge Alignment on Photo-Conversion Efficiency"
2024 MRS spring meeting & exhibit
Seattle, Washington

12-15 Dec 2023 — **Poster Presentation** "Emerging Van der Waals Heterostructures for Ultrathin Excitonic Solar Cells"
34th AGM of MRSI and 5th Indian Materials conclave
IIT BHU, India

24-28 July 2023 — **Poster Presentation** "Enhancing the Catalytic Activity of Zirconium Tri-Sulfides by Functionalization, Vacancy defect and Strain Engineering"
NAMMA Psi-k Workshop
JNCASR & IISC Banglore, India

2-4 Feb 2023 — **Poster Presentation(Best Poster)** "Enhancing the Catalytic Activity of Zirconium Tri-Sulfides by Functionalization, Vacancy defect and Strain Engineering"
Conference on Advances in Renewable Energy
HRI & TIFR Hyderabad, India

References

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